

SOUTH HATTERAS BUOY 41002, USA

WMO: 992150

Lat: 31.887N

Lon: 74.921W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 48.8	(c) 51.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 40.1	(k) 60.6	(l) 37.5	(m) 59.1	(n) 21.8	(o) 350	(p) 0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.7	(c) 84.1	(d) N/A	(e) 83.4	(f) N/A	(g) 83.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 12.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 34.5	(b) 31.7	(c) 29.0	(e) 44.2	(f) 85.2	(g) 4.0	(h) 1.2	(i) 41.3	(j) 86.1	(k) 38.9	(l) 86.7	(m) 36.7	(n) 87.4	(o) 33.8	(p) 88.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.8	63.6	63.6	64.9	68.6	72.4	77.3	80.2	80.3	79.0	74.9	69.6	66.9
	DBStd	7.24	5.73	5.28	5.07	3.20	2.62	2.53	1.69	1.65	2.01	3.30	4.29	5.09
	HDD50	4	1	1	2	0	0	0	0	0	0	0	0	0
	HDD65	297	95	76	62	9	0	0	0	0	0	1	13	41
	CDD50	7968	422	383	464	557	694	819	937	940	869	772	588	523
	CDD65	2787	51	38	60	116	229	369	472	475	419	308	151	99
	CDH74	18753	5	0	3	32	351	2676	4989	5061	3825	1534	235	42
	CDH80	2373	0	0	0	0	0	156	850	916	405	46	0	0
Wind	WSAvg	15.2	18.1	17.9	18.0	15.7	14.0	12.8	13.5	11.9	13.1	14.6	16.1	16.5
Precipitation	PrecAvg	57.0	5.5	5.2	4.8	3.3	3.3	4.2	4.8	5.2	6.1	5.0	4.8	4.8
	PrecMax	84.6	12.1	8.7	9.8	8.0	8.8	12.5	11.4	10.9	12.0	10.2	10.9	15.1
	PrecMin	43.1	2.0	1.5	1.3	1.1	1.1	0.5	0.8	1.9	1.3	1.4	1.7	1.4
	PrecStd	8.0	2.4	1.9	2.2	1.7	1.8	2.7	3.0	2.1	3.0	2.4	2.0	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.0	73.4	74.4	76.5	78.9	83.5	85.0	84.7	84.2	83.6	79.0	76.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	73.4	72.6	73.3	74.9	78.0	82.9	84.0	84.1	83.4	81.3	77.4	75.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	72.2	71.8	72.5	73.6	76.9	81.8	83.4	83.4	82.8	80.2	76.5	74.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	71.0	70.6	71.3	73.0	76.0	80.7	82.9	83.1	82.2	79.3	75.5	73.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.1	5.4	4.1	3.1	3.1	3.5	3.7	3.3	3.6	4.9	5.3
		MCDBR	6.4	5.8	6.0	4.6	3.3	3.1	3.5	3.8	3.6	3.7	4.9	5.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.361	0.370	0.382	0.407	0.439	0.451	0.459	0.448	0.420	0.394	0.368	0.364
	taud		2.505	2.484	2.445	2.358	2.312	2.339	2.345	2.378	2.449	2.484	2.511	2.503
	Ebn at Noon		272	281	285	281	271	266	264	266	270	269	268	265
	Edh at Noon		28	31	35	39	42	40	40	38	34	31	28	27
All-Sky Solar Radiation	RadAvg		793	1043	1414	1844	2067	2013	1868	1747	1532	1300	966	767
	RadStd		86	126	134	124	123	148	196	146	114	88	72	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page